

3D Printing as Material Probe: Supporting Fiber Understanding in Papermaking Craft

Beituo Liu
Tsinghua University
Beijing, China
1033940404@qq.com

Xuezhu Wang
Tsinghua University
Beijing, China
15626222995@163.com

Zhihao Yao
Tsinghua University
Beijing, Beijing, China
yaozh_h@outlook.com

Xiwen Yao
Tsinghua University
Beijing, China
xiiwen.yao@gmail.com

Yao Lu
Tsinghua University
Beijing, China
linaluyao@gmail.com

Haipeng Mi*
Tsinghua University
Beijing, China
mhp@tsinghua.edu.cn

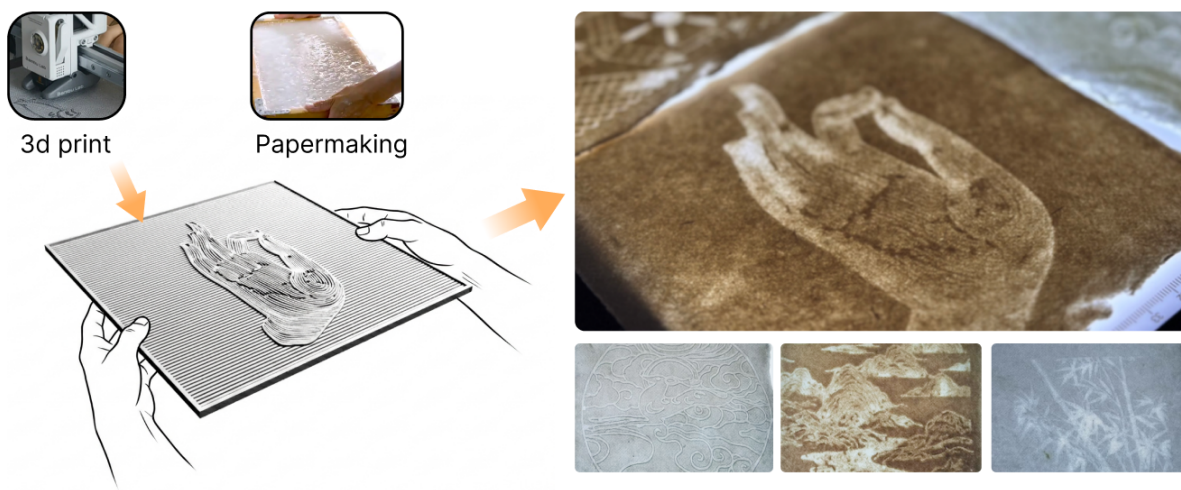


Figure 1: We used 3D printing to fabricate sheet-forming molds and explored a range of parameters and resulting artifacts with a focus on material behavior.

Abstract

Traditional crafts rely on tacit material knowledge that is difficult to articulate, teach, and iterate. In handmade papermaking, watermark design depends on subtle understanding of fiber behavior, drainage, and thickness variation—knowledge that is typically acquired through years of embodied practice. We present a material-centered fabrication workflow where 3D printing acts as a material probe to externalize this tacit knowledge. By using 3D-printed molds as interactive hypotheses about fiber behavior, craftspeople can systematically explore, observe, and reason about how material properties shape outcomes. Through field deployment in a traditional paper mill, we show how point-, line-, and surface-based

probes enabled craftspeople to see, compare, and discuss fiber behavior that was previously sensed only through experience. Our work reframes digital fabrication as a cognitive and material interaction tool rather than a means of automation, and contributes a transferable approach for augmenting material understanding in craft practices.

CCS Concepts

• **Human-centered computing** → **Interaction design**; *Empirical studies in HCI*.

Keywords

handmade paper, watermark design, digital fabrication, 3D printing, craft digitization, material-driven design

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* Corresponding author.



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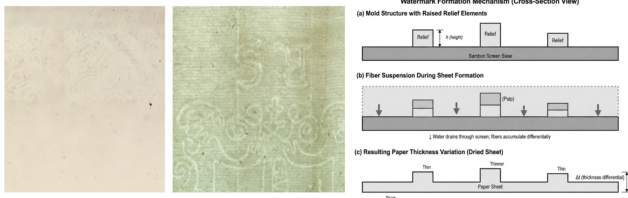


Figure 2: Ancient Chinese and European watermark papers (left) and the underlying mechanism of watermark formation (right).

1 Introduction

Traditional crafts rely on embodied material knowledge—skills learned through doing rather than explicit instruction. In handmade papermaking, watermark design depends on subtle relationships between fiber type, drainage, and mold geometry. Expert papermakers sense these relationships through experience, yet often struggle to articulate or systematically explore them. As a result, experimentation is slow and knowledge transfer relies heavily on long-term apprenticeship.

Digital fabrication is increasingly introduced to crafts to improve precision and efficiency. However, precision alone does not support material understanding. Parametric control can reproduce forms, but it rarely helps makers reason about why materials behave as they do. We argue that digital tools for craft should instead support material reasoning—helping makers explore, compare, and discuss material behavior through interaction.

In this paper, we present a material-centered fabrication approach where 3D printing acts as a material probe in papermaking. By externalizing hypotheses about fiber behavior into physical molds, craftspeople can observe outcomes, compare variations, and iteratively refine their understanding through making. We demonstrate this approach through field deployment in a traditional paper mill, showing how probe-based fabrication augments craft knowledge without disrupting embodied practice.

2 Related Work

2.1 Papermaking as Material Practice

Handmade papermaking is a material process in which form emerges from the interaction between fiber properties, water flow, and tool geometry. Watermarks are created through controlled thickness variation: raised elements on the mold locally reduce fiber accumulation, producing translucent patterns when backlit (Fig. 2). Historically, such elements were formed using woven threads or wire attached to bamboo screens, requiring craftspeople to rely on embodied experience to anticipate outcomes [1].

This practice depends on tacit knowledge. Papermakers learn to sense fiber behavior through resistance, timing, and visual cues, but these relationships are rarely articulated or systematically explored. As a result, experimentation is slow and knowledge transfer relies on long-term apprenticeship. While prior work has documented watermark history and preservation [4, 5], less attention has been paid to how craftspeople can “reason about” material behavior during making, or how tools might support this reasoning.

2.2 Material-Centered Interaction in HCI

HCI research has increasingly treated materials as active participants in interaction rather than passive substrates [9], emphasizing that understanding material behavior is central to creative practice [8, 10]. Recent work has shown how computational tools can augment material engagement in domains such as ceramics, textiles, carpentry [7], and lacquer art [2], while hybrid craft research highlights how digital fabrication can extend traditional practices without replacing embodied skill [3, 6]. However, most digital fabrication systems focus on producing form rather than supporting *material reasoning*. Building on this perspective, we reframe 3D printing as a material interaction medium: instead of automating papermaking, we use 3D-printed molds as material probes that make fiber behavior visible, comparable, and discussable during the craft process.

3 Watermark Formation Mechanism

The visibility of a watermark arises from differences in light transmittance across regions of a sheet of paper: thinner areas appear brighter under backlighting, while thicker areas appear darker. In handmade papermaking, a mold with raised geometric features is dipped into a fiber suspension. As water drains, fewer fibers settle on the raised regions, producing locally thinned, semi-transparent patterns.

This process is not controlled by a single parameter, but emerges from the interaction of material, mold geometry, and forming process.

- **Fiber material** determines watermark resolution. Long fibers (e.g., kozo) form visible fiber networks and provide high structural strength, but fine details are often masked by texture. Short fibers (e.g., cotton pulp) produce smoother surfaces and allow sharper edges and gradients, but become unstable under extreme geometric conditions.
- **Mold geometry** directly shapes fiber deposition. Traditional wire-stitched molds are limited by manual fabrication and struggle to achieve sub-millimeter features or continuous height variation. In contrast, FDM 3D-printed molds allow precise control of height, width, and gradients (± 0.1 mm), enabling watermarks to expand from linear symbols to continuous surfaces.
- **The forming process** amplifies or weakens these differences. Pulp consistency, lifting speed, and pressing conditions jointly determine whether stable thickness contrast can form at geometric boundaries.

Based on this understanding, we conceptualize watermarking as a material-aware geometric translation process: the mold does not directly “print” a pattern, but instead modulates the probability of fiber distribution, allowing optical differences to emerge within the paper.

4 Experimental Exploration

To translate the above mechanism into a designable parameter space, we conducted a three-stage experimental study that progressively narrowed down feasible geometric ranges and material choices (Fig. 3).

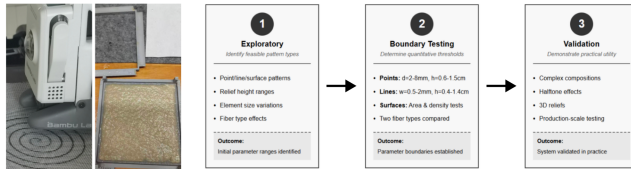


Figure 3: 3D printing setup (left) and the three-stage experimental workflow (right).

4.1 Phase 1: Exploratory Experiments

The first stage focused on open-ended exploration of pattern formation using four fiber substrates. We compared three sheet-forming screen types: a polypropylene mesh commonly used in contemporary handmade papermaking as a baseline, a 3D-printed mesh as a technical alternative, and a 3D-printed bamboo-inspired screen as a hybrid approach.

Through extensive fabrication of both traditional motifs and modern geometric patterns, we conducted close observation and comparative analysis of a large number of samples. Using a visual form-decomposition method, we distilled the diverse pattern vocabulary into three basic units: points, lines, and surfaces.

4.2 Stage 2: Threshold Identification

The second stage focused on systematically characterizing the three shape primitives identified in Stage 1. We conducted controlled parameter experiments using two representative fibers: unbleached cotton pulp (50% kozo + 50% wood pulp) and bleached kozo fiber. The color contrast between these materials made translucency differences easier to observe. A controlled-variable approach was used to identify the performance thresholds for each primitive.

- **Point primitives.** We tested circular dots (Fig. 5) and triangles, varying diameter, height, and distribution density. Circular forms gently guide pulp flow and support fiber aggregation, while polygonal forms introduce resistance and lead to distinct formation behaviors.
- **Line primitives.** We evaluated straight and curved lines by varying width, height, density, and gradient profiles. As the core element of traditional watermarking, lines carry a long craft lineage—from hemp-thread stitching in Chinese papermaking to copper wire molds in Western traditions. Our design (Fig. 6) preserves fine-line rhythms while extending them with controlled variation in spacing and relief.
- **Surface primitives.** Rarely used in traditional watermarking due to their impact on uniform formation, surface primitives are defined here as continuous-area geometries. Digital modeling and layered 3D printing enable such forms by decomposing surfaces into permeable structures. Using squares as representative samples (Fig. 7), we tested the limits of surface primitives across area and height dimensions.

4.3 Stage 3: Integrated Validation

The third stage followed two parallel directions: integrated validation and form innovation.

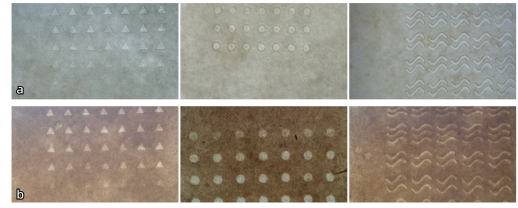


Figure 4: Preliminary variations of watermark patterns across two fiber materials. Samples made with bast fiber (paper mulberry) are shown in (a), and samples made with cotton pulp are shown in (b).

Table 1: Point Pattern Translucency Thresholds

Parameter	Min	Optimal	Max
Diameter (circles)	2mm	3–6mm	8mm
Diameter (triangles)	2mm	3–6mm	8mm
Height	0.6cm	0.8–1.2cm	1.5cm

Notes: <2mm invisible; triangles use circumscribed diameter; >1.5cm may tear.

In the composite pattern validation experiments, we expanded material testing to cotton pulp and kozo fiber, using the threshold data from previous stages to design patterns combining points, lines, and surfaces. We observed how different fiber characteristics affected visual outcomes, as well as the coordination and constraints among shape units.

In the relief watermark exploration, we extended watermarking beyond planar surfaces by integrating embossed geometry with sheet-forming functionality. Through precise digital modeling, we merged relief depth with screen structure to produce watermarks with spatial depth. Key technical challenges included height gradient control, demolding angles, and overall structural stability.

4.4 Results and Evaluation

The results show that 3D-printed molds not only improve watermark clarity but also expand the design language:

- Continuous height variation shifts watermarks from symbolic lines to tonal and relief expressions;
- Dot arrays and gradients enable halftone effects not possible with traditional wires;
- Curved and networked structures achieve sub-millimeter control while maintaining stability;
- Compared to traditional methods, fabrication time is reduced by approximately 30%, with significantly improved precision.

In validation tests conducted in a real papermaking workshop, molds were reusable for more than 50 cycles and required no changes to existing workflows. Craftspeople noted that the molds “make details possible that cannot be achieved with wire,” while still relying on material knowledge and embodied practice.

Diameter	Cotton Pulp		Mulberry Bark Pulp		Type	Cotton Pulp		Mulberry Bark Pulp	
	Circular Dots	Triangles	Circular Dots	Triangles		Straight Line	Curved Line	Straight Line	Curved Line
1mm/2mm/4mm					Width				
6mm					Height				
8mm					Density				

Figure 5: Integrated validation showing complex compositional applications of the established parameter system.

Table 2: Line Pattern Translucency Thresholds

Parameter	Min	Optimal	Max
Width	0.5mm	0.8–1.2mm	2mm
Height (cotton)	0.4cm	0.6–1.0cm	1.4cm
Height (mulberry)	0.6cm	0.8–1.2cm	1.4cm
Spacing	2mm	2–3mm	≥4mm

Notes: <0.5mm invisible; <2mm spacing blurs; cotton lower, mulberry higher threshold.

5 Ongoing Work

We plan to develop the translucency threshold system into an **interactive design tool** that generates manufacturable mold geometries from input graphics, grayscale images, or target translucency effects. The tool will support both forward and inverse design, allowing designers to explore outcomes from geometry or start from a desired visual result and infer height, width, and spacing parameters. Real-time feedback on material feasibility and failure risk will support efficient iteration, embedding material knowledge directly into the design process.

Beyond the forming stage, future work will explore **additional entry points across the papermaking process**, including drainage, pressing, and drying. For example, 3D-printed drainage structures could guide fiber deposition, and variable-geometry molds could modulate pressing textures. This approach aims to extend digital intervention from single-point control to process-wide coordination while preserving the character of handmade practice.

Finally, we will explore **material-level parameterization by mixing different fibers, pigments, functional particles, or biological materials into the pulp**. This would enable watermarks to express not only thickness variation, but also color, texture, and optical properties. Combined with the threshold system and design tool, material admixture expands watermarking from a single optical effect to a multi-material, multi-sensory expressive medium.

6 Conclusion

This research shows how digital fabrication can extend traditional watermark design through systematic parameter exploration and hybrid craft practice. By identifying translucency thresholds for

point, line, and surface elements, we externalized tacit craft knowledge into an explicit design framework. 3D printing enables geometric precision beyond traditional wire construction while remaining compatible with handmade papermaking, and production validation confirms that digital tools can augment rather than replace embodied skills. More broadly, this approach—combining parametric investigation, targeted digital fabrication, and collaboration with practitioners—offers a model for integrating digital technologies into traditional crafts, with future work exploring additional materials, design tools, fabrication methods, and contemporary applications.

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